



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India

Curriculum for Undergraduate Degree (B.Tech.) in Automobile Engineering (w.e.f. AY: 2020-21)

Part I: Introduction, Theme & Category wise Credit Distribution

A. Definition of Credit:

Sl. No.	Description	Credit
1	1 Hr. Lecture (L) Per Week	1.0
2	1 Hr. Tutorial (T) Per Week	1.0
3	1 Hr. Practical/ Lab (P) Per Week	0.5
4	2 Hrs. Practical/ Lab (P) Per Week	1.0

B. Range of Credits:

As per AICTE, a student covering 160 credits during 4 years of studies as per curriculum of the Institute will be eligible to get Under Graduate B.Tech. degree. A student will be eligible to get B.Tech. degree with Honours, if he/ she completes an additional 20 credits. These could be acquired through MOOCs prescribed by the Institute. Every student admitted to the 4 years B.Tech programme is required to earn a minimum of 100 Activity Points under Mandatory Additional Requirement (MAR), in addition to the required academic grades, for getting B.Tech./ B.Tech. (Honours) degree.

C. Category wise Credit Distribution:

Sl. No.	Category	Credit Allotted	Credit as per AICTE
1	Humanities and Social Sciences including Management Courses	12	12
2	Basic Science Courses	26	25
3	Engineering Science Courses including Workshop, Drawing, Basics of Electrical/ Mechanical/ Computer etc.	28	24
4	Professional Core Courses	57	48
5	Professional Elective Courses relevant to chosen specialization/ branch	15	18
6	Open Elective Courses from other technical and / or emerging subjects	9	18
7	Project Work, Seminar and Internship in Industry or elsewhere	13	15
8	Mandatory Courses [Environmental Sciences, Induction Training, Indian Constitution, Essence of Indian Traditional Knowledge, Aptitude Skill]	Non-Credit	Non-Credit
Total		160	160

D. Course Code and Definition:

Sl. No.	Course Code	Definitions
1	L	Lecture
2	T	Tutorial
3	P	Practical
4	BS	Basic Science Courses
5	ES	Engineering Science Courses
6	HM	Humanities and Social Sciences including Management Courses
7	PC	Professional Core Courses
8	PE	Professional Elective Courses
9	OE	Open Elective Courses
10	MC	Mandatory Courses
11	PW	Project/ Internships/ Sessional

E. Courses in different Category:

Humanities and Social Sciences including Management Courses							
Sl No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	I	HM-HU101	English	2	0	0	2
2	V	HM-HU501	Economics for Engineers	3	0	0	3
3	VI	HM-HU604	Quantitative Methods for Decision Making	3	0	0	3
4	VII	HM-HU704	Values & Ethics	2	0	0	2
Total Theory				9	0	0	10
Practical/ Sessional							
1	II	HM-HU291	Language Laboratory	0	0	2	1
2	V	HM-HU591	Soft Skill Development Laboratory	0	0	2	1
Total Practical/ Sessional				0	0	4	2
Total				9	0	4	12

Basic Science Courses							
Sl No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	I	BS-PH101	Physics	3	1	0	4
2	I	BS-M101	Mathematics-I	3	1	0	4
3	II	BS-CH201	Chemistry	3	1	0	4
4	II	BS-M201	Mathematics-II	3	1	0	4
5	III	BS-M303	Mathematics-III	2	1	0	3
6	III	BS-BIO301	Biology	2	0	0	2
7	IV	BS-M404	Numerical Methods	2	0	0	2
Total Theory				18	5	0	23
Practical/ Sessional							
1	I	BS-PH191	Physics Laboratory	0	0	3	1.5
2	II	BS-CH291	Chemistry Laboratory	0	0	3	1.5
Total Practical/ Sessional				0	0	6	3
Total				18	5	6	26

Engineering Science Courses including Workshop, Drawing, Basics of Electrical/ Mechanical/ Computer etc.							
Sl No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	I	ES-EE101	Basic Electrical and Electronics Engineering	3	1	0	4
2	II	ES-CS201	Programming for Problem Solving	3	0	0	3
3	III	ES-AUE301	Engineering Mechanics	3	1	0	4
4	III	ES-AUE302	Fluid Mechanics & Hydraulic Machines	4	0	0	4
5	IV	ES-AUE401	Materials Engineering	3	0	0	3
Total Theory				16	2	0	18
Practical/ Sessional							
1	I	ES-EE191	Basic Electrical & Electronics Engineering Lab	0	0	4	2
2	I	ES-ME191	Engineering Graphics & Design	1	0	4	3
3	II	ES-CS291	Programming for Problem Solving	0	0	4	2
4	II	ES-ME292	Workshop/Manufacturing Practices	1	0	4	3

Total Practical/ Sessional	2	0	14	10
Total	18	2	14	28

Professional Core Courses							
Sl No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	III	PC-AUE301	Applied Thermodynamics	3	1	0	4
2	III	PC-AUE302	Manufacturing Methods	4	0	0	4
3	IV	PC-AUE401	Strength of Materials	3	1	0	4
4	IV	PC-AUE402	Machine tools and Machining Technology	3	0	0	3
5	IV	PC-AUE403	Theory of Machine	3	1	0	4
6	IV	PC-AUE404	Metrology & Instrumentation	3	0	0	3
7	V	PC-AUE501	Automotive Engines	3	0	0	3
8	V	PC-AUE502	Automotive Body & Chassis Engineering	3	0	0	3
9	V	PC-AUE503	Heat Transfer	3	0	0	3
10	V	PC-AUE504	Design of Machine Element	3	1	0	4
11	VI	PC-AUE601	Automotive Transmission	3	0	0	3
12	VI	PC-AUE602	Automotive Electrical and Electronic Systems	3	0	0	3
13	VI	PC-AUE701	Vehicle Dynamics	3	0	0	3
Total Theory				40	5	0	44
Practical/ Sessional							
1	III	PC-AUE391	Machine Drawing	0	0	3	1.5
2	IV	PC-AUE491	Manufacturing &Testing Lab	0	0	3	1.5
3	V	PC-AUE591	Fluid Mechanics & Heat Transfer Lab	0	0	3	1.5
4	V	PC-AUE592	Automobile Engineering Lab I (Engine & Chassis Component Lab)	0	0	3	1.5
5	VI	PC-AUE691	Automobile Engineering Lab II (Engine Testing and Pollution Measurement Lab)	0	0	3	1.5
6	VI	PC-AUE692	Automobile Engineering Lab III (Automotive Design Lab)	0	0	3	1.5
7	VI	PC-AUE693	Automobile Engineering Lab IV (Vehicle Maintenance Lab)	0	0	3	1.5

8	VII	PC-AUE791	Automobile Engineering Lab V (Automotive Electrical & Electronics Lab)	0	0	3	1.5
9	VIII	PC-AUE881	Comprehensive Viva Voce	0	0	0	1
<i>Total Practical/ Sessional</i>				0	0	24	13
Total				40	5	24	57

Professional Elective Courses relevant to chosen specialization/ branch							
Sl No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	VI	PE-AUE601	Elective -I	3	0	0	3
2	VII	PE-AUE701	Elective -II	3	0	0	3
3	VII	PE-AUE702	Elective-III	3	0	0	3
4	VIII	PE-AUE801	Elective-IV	3	0	0	3
5	VIII	PE-AUE802	Elective-V	3	0	0	3
Total Theory				15	0	0	15
Practical/ Sessional							
1							
Total Practical/ Sessional				0	0	0	0
Total				15	0	0	15

List of Elective Courses	
PE-AUE601	Elective-I
PE-AUE601A	Hybrid & Electric Vehicles
PE-AUE601B	Transport Management & Motor Vehicles Act
PE-AUE601C	Tyre Technology
PE-AUE701	Elective – II
PE-AUE701A	Advanced Manufacturing Technology and Management
PE-AUE701B	Advanced Automotive Materials
PE-AUE701C	Finite Element Method & its Applications
PE-AUE702	Elective – III
PE-AUE702A	Automotive Component & System Design
PE-AUE702B	Automotive Pollution control & Alternate Fuels
PE-AUE702C	Two and Three Wheelers
PE-AUE801	Elective - IV
PE-AUE801A	Off Road Vehicles

PE-AUE801B	Automotive Air Conditioning
PE-AUE801C	Battery Technology and Management Systems
PE-AUE801D	Fuel cell Technology
PE-AUE802	Elective - V
PE-AUE802A	Thermal Management of Hybrid Systems
PE-AUE802B	Noise, Vibrations and Harshness
PE-AUE802C	Electric Motors for Automotive Applications

Open Elective Courses from other technical and / or emerging subjects							
Sl No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	VII		Open Elective – I	3	0	0	3
2	VIII		Open Elective - II	3	0	0	3
3	VIII		Open Elective - III	3	0	0	3
Total Theory				9	0	0	9
Practical/ Sessional							
1							
Total Practical/ Sessional				0	0	0	0
Total				9	0	0	9

List of Open Elective Courses	
Course Code	Open Elective – I
OE-ME701C	Quality Control & Reliability Engineering
OE-ME701B	Mechatronic Systems
OE-IT701C	Machine Learning
OE-CS701C	Cloud Computing
Course Code	Open Elective - II
OE-ME801F	Tribology
OE-IT801C	Internet of Things
OE-CS801H	Soft Computing
Course Code	Open Elective - III
OE-ME801B	Computational Fluid Dynamics
OE-HU801D	Entrepreneurship Development
OE-ME801E	Robotics and Robot Application
OE-ME801C	Non-Destructive Testing Methods

Project Work, Seminar and Internship in Industry or elsewhere							
Sl No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	V	PW-AUE581	Project-I	0	0	0	1
2	VI	PW-AUE671	Project-II	0	0	0	3
3	VII	PW-AUE781	Project-III	0	0	6	3
4	VIII	PW-AUE882	Project-IV	0	0	12	6
Total Theory				0	0	18	13
Practical/ Sessional							
1							
Total Practical/ Sessional				0	0	0	0
Total				0	0	18	13

Mandatory Courses [Environmental Sciences, Induction Training, Indian Constitution, Essence of Indian Traditional Knowledge, Aptitude Skill]							
Sl No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	IV	MC471	Environmental Science	2	0	0	0
2	V	MC571	Aptitude Skill Development-I	2	0	0	0
3	VI	MC671	Aptitude Skill Development-II	2	0	0	0
4	VI	MC673	Essence of Indian Knowledge Tradition	2	0	0	0
5	VII	MC772	Constitution of India	2	0	0	0
Total Theory				10	0	0	0
Practical/ Sessional							
Total Practical/ Sessional				0	0	0	0
Total				10	0	0	0

Curriculum for Under Graduate Degree (B.Tech.) in Automobile Engineering (w.e.f. AY: 2020-21)

First Semester (First Year)							
Mandatory Induction Program- 3 Weeks Duration							
Sl No.	Category	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	BS	BS-PH101	Physics	3	1	0	4
2	BS	BS-M101	Mathematics-I	3	1	0	4
3	ES	ES-EE101	Basic Electrical & Electronics Engineering	3	1	0	4
4	HM	HM-HU101	English	2	0	0	2
	Total Theory			11	3	0	14
Practical/ Sessional/ Mandatory Course							
1	BS	BS-PH191	Physics Laboratory	0	0	3	1.5
2	ES	ES-EE191	Basic Electrical & Electronics Engineering Lab	0	0	4	2
3	ES	ES-ME191	Engineering Graphics & Design	1	0	4	3
	Total Practical/ Sessional/ Mandatory Course			1	0	11	6.5
Total of First Semester				12	3	11	20.5

Second Semester (First Year)							
Sl No.	Category	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	BS	BS-CH201	Chemistry	3	1	0	4
2	BS	BS-M201	Mathematics–II	3	1	0	4
3	ES	ES-CS201	Programming for Problem Solving	3	0	0	3
Total Theory				9	2	0	11
Practical/ Sessional/ Mandatory Course							
1	BS	BS-CH291	Chemistry Laboratory	0	0	3	1.5
2	ES	ES-CS291	Programming for Problem Solving Lab	0	0	4	2
3	ES	ES-ME292	Workshop/ Manufacturing Practices	1	0	4	3
4	HM	HM-HU291	Language Laboratory	0	0	2	1
Total Practical/ Sessional/ Mandatory Course				1	0	13	7.5
Total of Second Semester				10	2	13	18.5

Third Semester (Second Year)							
Sl No	Category	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	BS	BS-M303	Mathematics -III	2	1	0	3
2	BS	BS-BIO301	Biology	2	0	0	2
3	ES	ES-AUE301	Engineering Mechanics	3	1	0	4
4	ES	ES-AUE302	Fluid Mechanics & Hydraulic Machines	4	0	0	4
5	PC	PC-AUE301	Applied Thermodynamics	3	1	0	4
6	PC	PC-AUE302	Manufacturing Methods	4	0	0	4
Total				18	3	0	21
Practical/ Sessional/ Mandatory Course							
1	PC	PC-AUE391	Machine Drawing	0	0	3	1.5
Total Practical/ Sessional/ Mandatory Course				0	0	3	1.5
Total of Third Semester				18	3	3	22.5

Fourth Semester (Second Year)							
Sl No	Category	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	BS	BS-M404	Numerical Methods	2	0	0	2
2	ES	ES-AUE401	Materials Engineering	3	0	0	3
3	PC	PC-AUE401	Strength Of Materials	3	1	0	4
4	PC	PC-AUE402	Machine Tools and Machining Technology	3	0	0	3
5	PC	PC-AUE403	Theory Of Machines	3	1	0	4
6	PC	PC-AUE404	Metrology & Instrumentation	3	0	0	3
Total				17	2	0	19
Practical/ Sessional/ Mandatory Course							
1	PC	PC-AUE491	Manufacturing &Testing Lab	0	0	3	1.5
2	MC	MC471	Environmental Sciences	2	0	0	0
Total Practical/ Sessional/ Mandatory Course				0	0	5	1.5
Total of Fourth Semester				19	2	3	20.5

Fifth Semester (Third Year)							
Sl No.	Category	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	PC	PC-AUE501	Automotive Engines	3	0	0	3
2	PC	PC-AUE502	Automotive Body & Chassis Engineering	3	0	0	3
3	PC	PC-AUE503	Heat Transfer	3	0	0	3
4	PC	PC-AUE504	Design of Machine Element	3	1	0	4
5	HM	HM-HU501	Economics for Engineers	3	0	0	3
Total Theory				15	1	0	16
Practical/ Sessional/ Mandatory Course							
1	PC	PC-AUE591	Fluid Mechanics & Heat Transfer Lab	0	0	3	1.5
2	PC	PC-AUE592	Automobile Engineering Lab-I (Engine & Chassis Component Lab)	0	0	3	1.5
3	HM	HM-HU591	Soft Skill Development Lab	0	0	2	1
4	PW	PW-AUE581	Project-I (Mini Project-30 hrs. Total)	0	0	2	1
5	MC	MC571	Aptitude Skill Development-I	2	0	0	0
Total Practical/ Sessional/ Mandatory Course				2	0	10	5
Total of Fifth Semester				17	1	10	21

Sixth Semester (Third Year)							
Sl No.	Category	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	PC	PC-AUE601	Automotive Transmission	3	0	0	3
2	PC	PC-AUE602	Automotive Electrical and Electronic Systems	3	0	0	3
3	PE	PE-AUE601	Professional Elective-I	3	0	0	3
4	HM	HM-HU604	Quantitative Methods for Decision Making	3	0	0	3
Total Theory				12	0	0	12
Practical/ Sessional/ Mandatory Course							
1	PC	PC-AUE691	Automobile Engineering Lab II (Engine Testing And Pollution Measurement Lab)	0	0	3	1.5
2	PC	PC-AUE692	Automobile Engineering Lab III (Automotive Design Lab)	0	0	3	1.5
3	PC	PC-AUE693	Automobile Engineering Lab IV (Vehicle Maintenance Lab)	0	0	3	1.5
4	MC	MC671	Aptitude Skill Development-II	2	0	0	0
5	MC	MC673	Essence Of Indian Knowledge Tradition	2	0	0	0
6	PW	PW-AUE671	Project-II (Summer Internship-90 Hrs. Total)	0	0	0	3
Total Practical/ Sessional/ Mandatory Course				4	0	9	7.5
Total of Sixth Semester				16	0	9	19.5

List of Elective Courses	
PE-AUE601	Professional Elective-I
PE-AUE601A	Hybrid & Electric Vehicles
PE-AUE601B	Transport Management & Motor Vehicles Act
PE-AUE601C	Tyre Technology

Seventh Semester (Fourth Year)							
Sl No.	Category	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	PC	PC-AUE701	Vehicle Dynamics	3	0	0	3
2	PE	PE-AUE701	Professional Elective-II	3	0	0	3
3	PE	PE-AUE702	Professional Elective-III	3	0	0	3
4	OE		Open Elective-I	3	0	0	3
5	HM	HM-HU704	Values & Ethics	2	0	0	2
Total				14	0	0	14
Practical/ Sessional/ Mandatory Course							
1	PC	PC-AUE791	Automobile Engineering Lab V (Automotive Electrical & Electronics Lab)	0	0	3	1.5
2	PW	PW-AUE781	Project-III	0	0	6	3
3	MC	MC772	Constitution of India	2	0	0	0
Total Practical/ Sessional/ Mandatory Course				0	0	9	4.5
Total of Seventh Semester				14	0	9	18.5

List of Elective Courses	
PE-AUE701	Professional Elective-II
PE-AUE701A	Advanced Manufacturing Technology and Management
PE-AUE701B	Advanced Automotive Materials
PE-AUE701C	Finite Element Method & its Applications
PE-AUE702	Professional Elective-III
PE-AUE702A	Automotive Component & System Design
PE-AUE702B	Automotive Pollution control & Alternate Fuels
PE-AUE702C	Two and Three Wheelers
List of Open Elective Courses	
Open Elective-I	
OE-ME701C	Quality Control & Reliability Engineering
OE-ME701B	Mechatronic Systems
OE-IT701C	Machine Learning
OE-CS701C	Cloud Computing

Eighth Semester (Fourth Year)							
Sl No.	Category	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	PE	PE-AUE801	Professional Elective-IV	3	0	0	3
2	PE	PE-AUE802	Professional Elective-V	3	0	0	3
3	OE		Open Elective-II	3	0	0	3
4	OE		Open Elective-III	3	0	0	3
Total Theory				12	0	0	12
Practical/ Sessional/ Mandatory Course							
1	PC	PC-AUE881	Comprehensive Viva Voce	0	0	0	1
2	PW	PW-AUE882	Project-IV	0	0	12	6
Total Practical/ Sessional/ Mandatory Course				0	0	12	7
Total of Eighth Semester				12	0	12	19

List of Elective Courses	
PE-AUE801	Professional Elective-IV
PE-AUE801A	Off Road Vehicles
PE-AUE801B	Automotive Air Conditioning
PE-AUE801C	Battery Technology and Management Systems
PE-AUE801D	Fuel cell Technology
PE-AUE802	Professional Elective-V
PE-AUE802A	Thermal Management of Hybrid Systems
PE-AUE802B	Noise, Vibrations and Harshness
PE-AUE802C	Electric Motors for Automotive Applications
List of Open Elective Courses	
Open Elective-II	
OE-ME801F	Tribology
OE-IT801C	Internet of Things
OE-CS801H	Soft Computing
Open Elective-III	
OE-ME801B	Computational Fluid Dynamics
OE-HU801D	Entrepreneurship Development
OE-ME801E	Robotics and Robot Application
OE-ME801C	Non-Destructive Testing Methods